

Raspberry Pi Installation and Configuration Documentation

Screen installation on the Raspberry PI

We have been given these components .



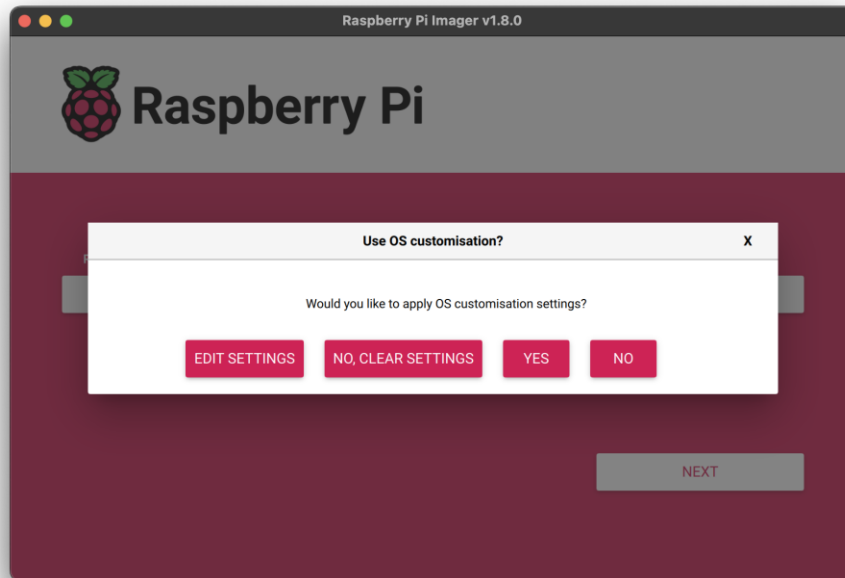
We must connect the raspberry pi to the screen



We should also screw the raspberry PI to the back of the screen

Installing Raspberry Pi OS

I downloaded the Raspberry Pi Imager and selected Raspberry Pi OS (64-bit). I configured the hostname in the installation screen



Basic System Configuration

After booting, I opened a terminal and updated the system.

```
a@allah657:~$ sudo apt update
```

Then I set up the screen orientation to landscape mode.



Static IP and Hostname Configuration

I set the hostname to allah657 and configured a static IP (10.12.2.42).

```
interface eth0
static ip_address=10.12.2.42/24
```

SSH configuration

To be able to ssh into it without a password then we need to generate a key

To generate key on your computer:

```
ssh-keygen -t ed25519
```

Copy key to Raspberry Pi:

```
ssh-copy-id a@allah657
```

```
cat ~/.ssh/id_ed25519.pub | ssh pi@raspberrypi.local "mkdir -p ~/.ssh && cat >>
~/.ssh/authorized_keys"
```

Test login:

```
a@allah657
```

To disable password login on the Pi:

Edit:

```
sudo nano /etc/ssh/sshd_config
```

Set:

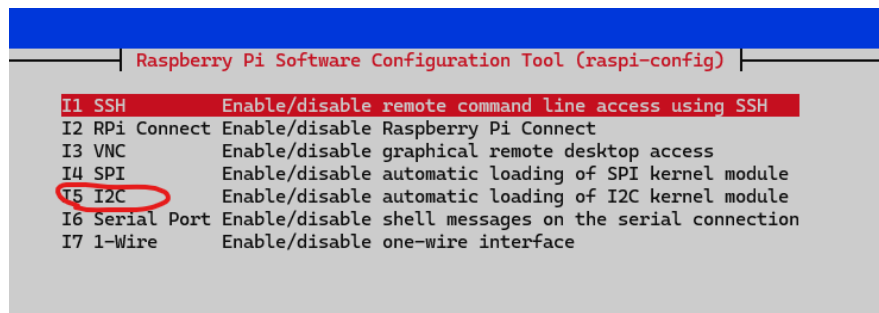
```
PasswordAuthentication no
```

Restart SSH:

```
sudo systemctl restart ssh
```

System configuration.

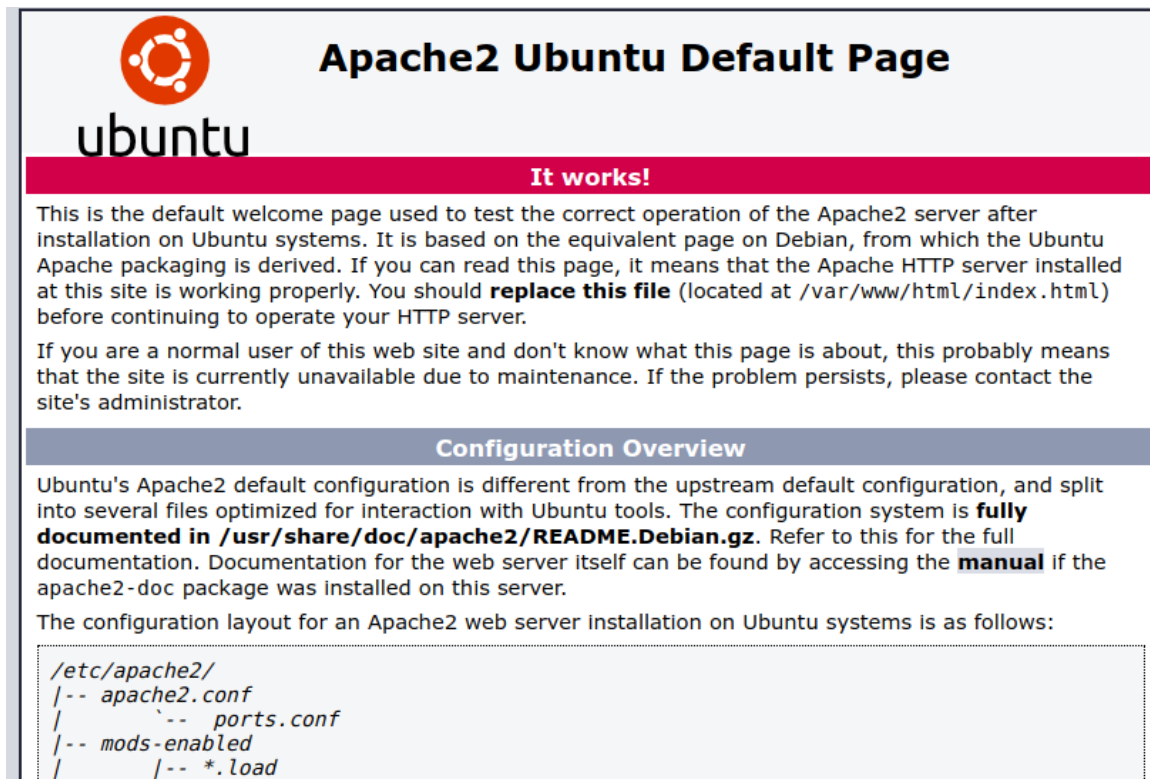
I enabled the I²C interface through raspi-config.



I have also enabled VNC from the same screen.

Installing and configuring docker and Apache

To install Apache: "sudo apt install apache2"



The screenshot shows the Apache2 Ubuntu Default Page. At the top left is the Ubuntu logo. To its right is the title "Apache2 Ubuntu Default Page". Below the logo is the word "ubuntu". A red banner with the text "It works!" is centered. Below this, a paragraph explains that this is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It mentions that the page is based on the equivalent page on Debian and that if the user can read this page, it means the Apache HTTP server is working properly. It advises replacing the file at /var/www/html/index.html before continuing to operate the HTTP server. Another paragraph explains that if a normal user doesn't know what the page is about, it probably means the site is currently unavailable due to maintenance. Below this is a section titled "Configuration Overview" in a blue header. The text explains that Ubuntu's Apache2 default configuration is different from the upstream default and is split into several files optimized for interaction with Ubuntu tools. It states that the configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz and refers to this for full documentation. It also mentions that documentation for the web server itself can be found by accessing the manual if the apache2-doc package was installed. Finally, it states that the configuration layout for an Apache2 web server installation on Ubuntu systems is as follows: /etc/apache2/. Below this text is a code block showing the directory structure: /etc/apache2/, |-- apache2.conf, | |-- ports.conf, |-- mods-enabled, | |-- *.load.

I installed Docker using the official get.docker.com script.

```
curl -fsSL https://get.docker.com -o install-docker.sh
```

then i ran the script to install docker on my machine.

Creating Required Folders

I created /opt/station, /var/station, and /tmp/station with correct permissions

I took ownership over each folder "chown a:a FOLDERNAME" "a" is my username

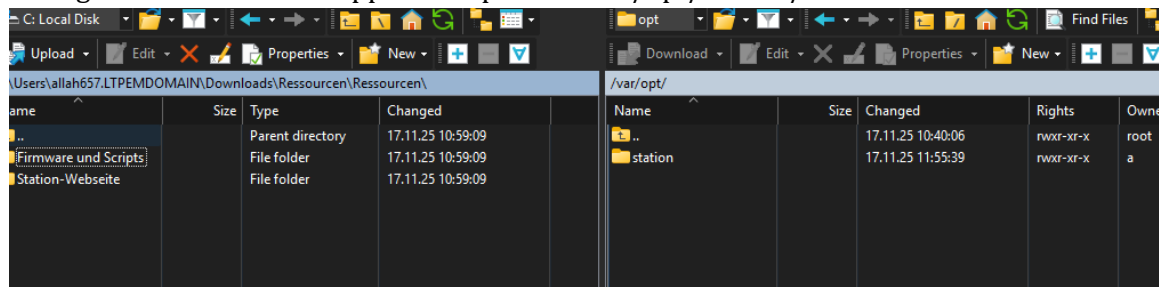
I have also given both folders chmod 777 just so I don't run into any permission problems.

I know it is bad practice but it works.

Copying Supplied Files

I copied the firmware files, website files, and sensor test scripts to their respective directories.

We had a problem where the documentation would say copy the website files into the wrong folder and we are supposed to put them in /opt/station/www



Creating the Docker Container

I created a PHP-Apache container and confirmed it was running.

```
docker run -d --name php -p 8080:80 -v /tmp/station:/tmp/station -v /opt/station/www:/var/www/html php:8.4-apache
```

```
Warning: file(/tmp/station/result.csv): Failed to open stream: No such file or directory in /var/www/html/index.php on line 3
Time
Temperature      0.00 °C
Humidity         0.00 %
Pressure         0.00 hPa
Light            0.00 lux
Gas              0.00 ppm
```

Autostart Setup

We were asked to make the website host and launch the browser into fullscreen mode, so that when you turn on the raspberry pi it would go to that website automatically so

I configured ~/.config/labwc/autostart so the firmware, Docker container and browser start automatically

```
#!/bin/bash
# Create necessary directories
mkdir -p /tmp/station
# Start Docker container with website from /var/www/html
sudo docker start php || docker run -d --name php -p 8080:80 -v /tmp/station:/tmp/station -v /var/www/html:/var/www/html
# Wait for services to start
sleep 5
# Start measurement script using virtual environment
cd /opt/station
source venv/bin/activate
python3 measure.py &
# Wait for measurements to initialize
sleep 10
# Start browser in kiosk mode (fullscreen)
chromium --kiosk http://localhost:8080 &
```